Vo Van Tu Tai

Full-Stack Developer · PHP & Modern Web Systems · AI-Driven Engineering

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PROFESSIONAL SUMMARY

Information Systems Engineer (IUH, GPA 3.26/4.0 · Completed all degree requirements · Available immediately for full-time) with hands-on full-stack development experience spanning **PHP (OOP & MVC), JavaScript/TypeScript, Node.js (Express, NestJS), and React**. Proficient in relational data modeling (**MySQL, PostgreSQL 3NF**), RESTful API contracts (OpenAPI 3.0), and Docker containerization. Active practitioner of **AI-Driven Development**, effectively leveraging **Claude Code, Cursor, and GitHub Copilot** to accelerate feature delivery, refactor clean code, and author test suites. Possesses strong analytical thinking, structured problem-solving skills, and high motivation to master scalable, high-performance web architectures and clean code design patterns.

CORE TECHNICAL COMPETENCIES

Programming Languages: JavaScript (ES6+), TypeScript, PHP (OOP, MVC), Python, Java (Core OOP), SQL, HTML5, CSS3.

Frontend Technologies: React.js, Responsive Web Design, Modern Vanilla CSS, Bootstrap, DOM Manipulation, Single Page Applications (SPA).

Backend & API Design: Node.js (Express.js, NestJS), PHP (Custom MVC Architecture), RESTful API Design (OpenAPI 3.0, Postman), Asynchronous Webhooks, RabbitMQ Message Broker, Concurrency Control.

Databases & Storage: MySQL, PostgreSQL, MongoDB, Redis (In-memory caching & locks), Relational Database Design (ERD 3NF), SQL Query Optimization.

AI-Driven Development: Claude Code, GitHub Copilot, Cursor, ChatGPT (AI-assisted rapid prototyping, code refactoring, test case generation, API contract synthesis).

DevOps, Cloud & Tools: Docker, Docker Compose, Git / GitHub, Postman, Linux / Bash fundamentals, Cloud Exposure (AWS / GCP basics), Jira, Figma.

Engineering Practices: MVC & OOP Design Patterns, Data Structures & Algorithms, Clean Code, Agile / Scrum Methodology, Cross-functional Collaboration.

WORK EXPERIENCE

HomeNest Software
Ho Chi Minh City, Vietnam

Business Analyst · PM Assistant (Engineering Liaison)
Mar 2026 – Sep 2026

- **Technical Specification & Schema Alignment:** Collaborated closely with Lead Dev and UI/UX to translate multi-service business logic (On-demand home repair, carpooling, marketplace) into precise technical specifications, JSON data models, and testable acceptance criteria (Given-When-Then), preventing schema mismatches before sprint kickoff.
- **State-Machine Architecture (Interactive Bidding):** Architected a state-driven price negotiation module (propose ↔ counter-bid ↔ accept/reject with TTL expiration); specified strict status locking and race-condition guards to guide backend developers in implementation.
- **API & Performance Cost Optimization:** Resolved high third-party map API budget consumption and mobile battery drain by specifying an event-driven GPS refresh checkpoint mechanism (triggered strictly on order pickup/completion instead of continuous polling), drastically saving external API invocation costs while ensuring defensible audit trails.
- **QA Testing & UAT Coordination:** Executed hands-on functional and smoke testing on staging builds using Postman and web clients; conducted defect triage in Jira, coordinating rapid bug resolution with developers across 2+ release cycles.

J&T Express
Ho Chi Minh City, Vietnam

Operations Process Coordinator (Full-time)
Sep 2025 – Feb 2026

- **SQL Data Reconciliation & System Integrity:** Formulated daily SQL queries to cross-verify discrepancies between physical branch delivery manifests and enterprise logistics system logs; prepared daily SQL data integrity reports for management.
- **Pipeline Throughput Optimization:** Identified throughput bottlenecks handling high-volume daily shipments; restructured shift processing into a closed-loop unidirectional sorting flow (Intake → Weigh → Scan → Dispatch), substantially accelerating turnaround times.
- **Standard Operating Procedures (SOP):** Authored standard operational exception procedures categorizing delivery anomalies, establishing clear boundary checks and cross-shift operational accountability.

FEATURED TECHNICAL PROJECTS

Song Tai Shop — Full-Stack E-Commerce Platform [GitHub ↗] Web Application Project

PHP (OOP, Custom MVC Framework), MySQL (3NF & ACID), JavaScript (Fetch API/AJAX), Docker Compose *Nov 2025 – Jan 2026*

- **Custom MVC Architecture & Security:** Engineered a native PHP MVC framework from scratch utilizing a Front-Controller (`index.php`), custom regex routing for REST-like methods, PDO prepared statements eliminating SQL injection, CSRF token validation, XSS sanitization, and session-based RBAC.
- **ACID Transactional Cart & Concurrency:** Protected inventory integrity during simultaneous checkouts by wrapping order placement and stock deduction in strict MySQL ACID transactions (`START TRANSACTION`, `COMMIT`, `ROLLBACK`) with pessimistic row locking (`SELECT ... FOR UPDATE`).
- **Dynamic Client UI & Containerization:** Developed dynamic shopping cart interactions using JavaScript Fetch API for real-time quantity recalculation without page reloads; packaged the full LAMP stack (PHP-FPM, Apache, MySQL) into **Docker Compose** for reproducible zero-friction local deployments.

Tourist Accommodation Booking & Hospitality Platform [GitHub ↗] Web Application Project

PHP (OOP & MVC Architecture), MySQL (3NF & Indexing), JavaScript, VNPay IPN Webhook, RESTful APIs, Docker *Oct 2025 – Dec 2025*

- **Full-Stack PHP Booking Engine & RBAC:** Architected an end-to-end multi-tenant booking platform in **PHP OOP & MVC** with RESTful APIs for catalog search, date-range availability checks, and multi-tier pricing; enforced granular RBAC isolating Guests, Hosts, and Super-Admins.
- **Concurrency Defense & Overbooking Lock (TTL Hold):** Solved critical room reservation race conditions during peak traffic spikes through a stateful two-phase booking hold: temporarily locks rooms in `Pending` state with a 15-minute TTL hold, auto-releasing inventory via timeout daemon if unconfirmed.
- **Cryptographic Payment Gateway & Indexing:** Integrated VNPay payment gateway using HMAC-SHA512 checksum validation with idempotent IPN webhook processing; optimized MySQL 3NF database with composite indexes on `(property_id, check_in, check_out, status)` for sub-second query latency.

Nexus Express — Logistics OMS & Distributed Workflow Engine [Live Demo ↗] [GitHub ↗] Distributed Architecture Project

NestJS, TypeScript, PostgreSQL (3NF), RabbitMQ, React, OpenAPI 3.0, Camunda BPMN 2.0, Docker *Sep 2025 – Feb 2026*

- **Asynchronous Event-Driven Microservices & Live Tracking:** Built an enterprise Order Management System with live tracking deployed at **tracking.nexus-ex.site** using **NestJS (TypeScript)** and **RabbitMQ**, establishing decoupled message queues for dispatch and waypoint tracking to absorb high burst traffic without dropped messages.
- **Idempotent API Contracts & Camunda BPMN:** Designed OpenAPI 3.0 specifications enforcing `Idempotency`–Key headers on scan endpoints to prevent duplicate records upon retries; modeled parcel lifecycles with Camunda BPMN 2.0 to enforce strict transition validation gates.
- **PostgreSQL 3NF Data Modeling & Audit Trails:** Designed normalized schema with composite B-Tree indexes on tracking codes and timestamps; maintained tamper-proof event audit logs enabling 100% discrepancy reconciliation during transit audits.

Cab Booking & Real-Time Dispatch Platform [GitHub ↗] High-Concurrency Distributed Project

Node.js, Express.js, Redis (In-Memory Caching & Distributed Locks), Leaflet / Geolocation, Docker Compose *Dec 2025 – Jan 2026*

- **High-Concurrency Dispatch with Redis Locks:** Engineered driver-rider matching backend utilizing **Redis distributed locks (SETNX with TTL)**, completely eliminating simultaneous ride acceptance race conditions when multiple drivers accept the same trip.
- **In-Memory Caching & Geospatial Queries:** Leveraged Redis geospatial indexes (`GEOADD`, `GEORADIUS`) to track real-time driver coordinates and calculate proximity in sub-milliseconds, bypassing expensive disk-based database lookups.
- **State Machine Resiliency & Containerization:** Modeled resilient ride state machine (`Requested → Accepted → Arrived → In–Trip → Completed / Cancelled`) with dynamic surge pricing; containerized services, Redis cache, and databases into Docker Compose with health checks.

EDUCATION

Industrial University of Ho Chi Minh City (IUH) Ho Chi Minh City, Vietnam

Bachelor of Engineering (B.Eng.) in Information Systems · GPA: 3.26 / 4.0 *Sep 2022 – 2026 (Completed All Degree Requirements · Available Full-time)*

- **Relevant Coursework:** Software Engineering, Object-Oriented Programming (OOP), Data Structures & Algorithms, Systems Analysis & Design, Database Management Systems (Advanced SQL Queries & Indexing), Web Application Development, Service-Oriented Architecture (SOA), Enterprise Resource Planning (ERP - Odoo), Artificial Intelligence & Intelligent Systems Concepts.
- **Languages & Professional Traits:** English (Professional Working Proficiency — capable of reading English technical documentation, API specifications & developer discussions), Vietnamese (Native), proactive problem-solving, and rapid adaptation to modern engineering workflows.